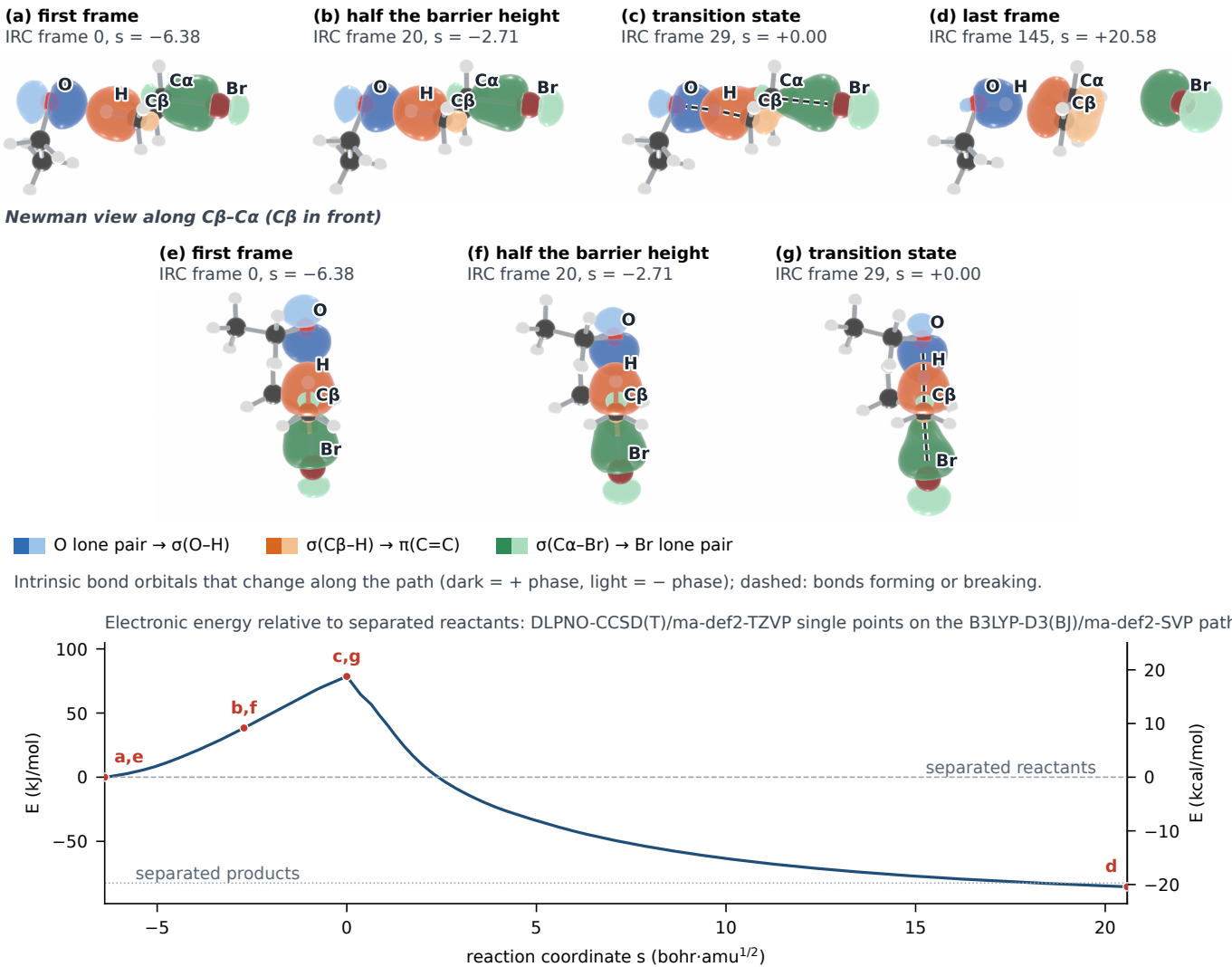

(a) first frame
IRC frame 0, s = −6.38
(b) half the barrier height
IRC frame 20, s = −2.71
(c) transition state
IRC frame 29, s = +0.00
(d) last frame
IRC frame 145, s = +20.58
O
H
Cα
Cβ
Br
Newman view along Cβ–Cα (Cβ in front)
(e) first frame
IRC frame 0, s = −6.38
(f) half the barrier height
IRC frame 20, s = −2.71
(g) transition state
IRC frame 29, s = +0.00
O lone pair → σ(O–H)
σ(Cβ–H) → π(C=C)
σ(Cα–Br) → Br lone pair
Intrinsic bond orbitals that change along the path (dark = + phase, light = − phase); dashed: bonds forming or breaking.
Electronic energy relative to separated reactants: DLPNO-CCSD(T)/ma-def2-TZVP single points on the B3LYP-D3(BJ)/ma-def2-SVP path
E (kJ/mol)
E (kcal/mol)
a,e
b,f
c,g
d
separated reactants
separated products
reaction coordinate s (bohr·amu^1/2)